

Linear differential equations in $\mathbb{R}^n$

1. LHDE in $\mathbb{R}^2$ with CC

We consider a system with 2 equations and 2 unknowns

$$(1) \begin{cases} x' = a_{11}x + a_{12}y \\ y' = a_{21}x + a_{22}y \end{cases} \quad \text{where} \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \in M_2(\mathbb{R}).$$

For $X = \begin{pmatrix} x \\ y \end{pmatrix}$: (1) $\Leftrightarrow$ $X' = AX$

Def. A solution of (1) is a function $\varphi: \mathbb{R} \rightarrow \mathbb{R}^2$, of class C^1 s.t.
 $\varphi'(t) = A\varphi(t)$, $\forall t \in \mathbb{R}$.

We say that (1) is uncoupled when $a_{12} = a_{21} = 0$. $\begin{cases} x' = a_{11}x \\ y' = a_{22}y \end{cases}$
 otherwise, we say that (1) is coupled.

Prop. The gen. sol. of an uncoupled system is $\begin{cases} x = c_1 e^{a_{11}t} \\ y = c_2 e^{a_{22}t} \end{cases}$, $c_1, c_2 \in \mathbb{R}$.

From now on, we consider that either $a_{12} \neq 0$ or $a_{21} \neq 0$.

Remark. For a second order LHDE $x'' + b_1x' + b_2x = 0$ ($b_1, b_2 \in \mathbb{R}$),
 we denote $y = x'$ and we arrive to a LHDE in $\mathbb{R}^2$:

$$\begin{cases} x' = y \\ y' = -b_2x - b_1y \end{cases} \quad \left(y' = x'' = -b_1x' - b_2x = -b_1y - b_2x \right)$$

For this system, the coefficients matrix is $A = \begin{pmatrix} 0 & 1 \\ -b_2 & -b_1 \end{pmatrix}$.

The reduction method to find the general solution of a coupled (1).

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$$x' = a_{11}x + a_{12}y$$

Our aim is
to find
a relation
 x'' , x' and x .

Step 1. $y = \frac{1}{a_{12}} (x' - a_{11}x)$

Step 2. $x'' = a_{11}x' + a_{12}y'$

Step 3. We use step 2 and the second eq. $y' = a_{21}x + a_{22}y$

$$x'' = a_{11}x' + a_{12}a_{21}x + a_{12}a_{22}y$$

Step 4. We use Step 1 and Step 3

$$x'' = a_{11}x' + a_{12}a_{21}x + a_{12}a_{22} \cdot \frac{1}{a_{12}} (x' - a_{11}x)$$

$$x'' - (a_{11} + a_{22})x' + (a_{11}a_{22} - a_{12}a_{21})x = 0 \quad (2)$$

Step 5. We find the gen. sol. of (2) $x = c_1 x_1 + c_2 x_2$

Step 6. We find y using Step 1 and Step 5.

Example. Find the gen. sol. of $\begin{cases} x' = -y \\ y' = 4x \end{cases} \quad A = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix}$

$$y = -x'$$

$$x'' = -y' \quad y' = 4x$$

$$x'' = -4x \Rightarrow x'' + 4x = 0 \quad r^2 + 4 = 0 \quad r^2 = -4 \quad r_{1,2} = \pm 2i$$

$$\hookrightarrow \cos 2t, \sin 2t \quad x = c_1 \cos 2t + c_2 \sin 2t, \quad c_1, c_2 \in \mathbb{R}$$

$$y = -x' = -(-2c_1 \sin 2t + 2c_2 \cos 2t)$$

The general sol. is $\begin{cases} x = c_1 \cos 2t + c_2 \sin 2t \\ (3) \quad y = -2c_2 \cos 2t + 2c_1 \sin 2t \end{cases}, \quad c_1, c_2 \in \mathbb{R}.$

Exercise. Find the sol. of the IVP's

$$a) \quad x' = -y, \quad y' = 4x, \quad x(0) = 1, \quad y(0) = 0. \quad \begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$b) \quad x' = -y, \quad y' = 4x, \quad x(0) = 0, \quad y(0) = 1. \quad \begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

pt In the gen. sol. (3) we plug $t=0$: $x(0) = c_1, \quad y(0) = -2c_2.$

a) $c_1 = 1, \quad c_2 = 0 \Rightarrow$ This IVP has a unique solution

$$\varphi_a: \mathbb{R} \rightarrow \mathbb{R}^2, \quad \varphi_a(t) = \begin{pmatrix} \cos 2t \\ 2 \sin 2t \end{pmatrix}, \quad t \in \mathbb{R}.$$

b) $c_1 = 0, \quad c_2 = -\frac{1}{2} \Rightarrow$ This IVP has a unique sol.

$$\varphi_b: \mathbb{R} \rightarrow \mathbb{R}^2, \quad \varphi_b(t) = \begin{pmatrix} -\frac{1}{2} \sin 2t \\ \cos 2t \end{pmatrix}, \quad t \in \mathbb{R}.$$

Exercise. Consider $E(t) = \begin{pmatrix} \cos 2t & -\frac{1}{2} \sin 2t \\ 2 \sin 2t & \cos 2t \end{pmatrix}, \quad A = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix}$

Check that $E'(t) = A E(t), \quad t \in \mathbb{R}$ and $E(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$

Find the eigenvalues of A .

Proof. it is easy to see that $E(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$

$$E'(t) = \begin{pmatrix} -2 \sin 2t & -\cos 2t \\ 4 \cos 2t & -2 \sin 2t \end{pmatrix}$$

$$A E(t) = \begin{pmatrix} 0 & -1 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} \cos 2t & -\frac{1}{2} \sin 2t \\ 2 \sin 2t & \cos 2t \end{pmatrix} = \begin{pmatrix} -2 \sin 2t & -\cos 2t \\ 4 \cos 2t & -2 \sin 2t \end{pmatrix}$$

$$\Rightarrow E'(t) = A E(t), \quad t \in \mathbb{R}.$$

For the eigenvalues of A : $\lambda^2 - (\text{tr} A) \lambda + \det A = 0$

$$\text{tr} A = 0, \quad \det A = 4 \quad \lambda^2 + 4 = 0 \quad \lambda_{1,2} = \pm 2i$$

$$\sigma(A) = \{-2i, 2i\}.$$

Proposition. The characteristic equation of (2) is

Proposition. The characteristic equation of (2) is

$$r^2 - (\text{tr} A)r + \det A = 0, \text{ where } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

is the coefficient's matrix of system (1).

Thus, the roots of the characteristic eq. of (2) are the eigenvalues of A .

Definition. The principal matrix solution of system (1): $x' = Ax$

is a C^1 function $E: \mathbb{R} \rightarrow M_2(\mathbb{R})$ such that

$$E'(t) = A E(t), \quad t \in \mathbb{R} \quad \text{and} \quad E(0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Prop. Let $\varphi_a: \mathbb{R} \rightarrow \mathbb{R}^2$ be the sol. of the IVP $x' = Ax, x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and let $\varphi_b: \mathbb{R} \rightarrow \mathbb{R}^2$ be the sol. of the IVP $x' = Ax, x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Then $E: \mathbb{R} \rightarrow M_2(\mathbb{R}), E(t) = (\varphi_a(t) \quad \varphi_b(t)), t \in \mathbb{R}$ is the principal matrix sol. of $x' = Ax$.

Proof. $\varphi_a'(t) = A \varphi_a(t), \quad \varphi_a(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$\varphi_b'(t) = A \varphi_b(t), \quad \varphi_b(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow E(0) = (\varphi_a(0) \quad \varphi_b(0)) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \checkmark$$

$$\left. \begin{aligned} E'(t) &= (\varphi_a'(t) \quad \varphi_b'(t)) = (A \varphi_a(t) \quad A \varphi_b(t)) \\ A E(t) &= A \cdot (\varphi_a(t) \quad \varphi_b(t)) = (A \varphi_a(t) \quad A \varphi_b(t)) \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow E'(t) = A E(t), \quad t \in \mathbb{R}.$$

Theorem. For any $\eta \in \mathbb{R}^2$, the IVP $x' = Ax, x(0) = \eta$

has a unique solution and this is $\boxed{\varphi(t, \eta) = E(t)\eta}$,
 where $E(t)$ is the principal matrix sol. of $X' = AX$.

Proof. $\varphi(t) = E(t)\eta$. We want to check that

$$\varphi'(t) = A\varphi(t), \quad t \in \mathbb{R} \quad \text{and} \quad \varphi(0) = \eta.$$

$$\text{Indeed, } \varphi(0) = E(0)\eta = I_2\eta = \eta \quad \checkmark$$

$$\varphi'(t) = E'(t)\eta = AE(t)\eta = A\varphi(t) \quad \checkmark.$$

How to find directly $E(t)$?

$$? \quad E'(t) = aE(t), \quad E(0) = 1 \quad (a \in \mathbb{R})$$

$$E(t) = e^{at}, \quad t \in \mathbb{R}.$$

The matrix exponential

$$\text{For } a \in \mathbb{R} \text{ we have } e^a = \sum_{k=0}^{\infty} \frac{1}{k!} a^k = 1 + a + \frac{1}{2!}a^2 + \frac{1}{3!}a^3 + \dots$$

absolutely convergent.

$$\text{For } z \in \mathbb{C} \text{ we have } e^z = \sum_{k=0}^{\infty} \frac{1}{k!} z^k \text{ is absolutely convergent.}$$

in addition, for any matrix $A \in \mathcal{M}_2(\mathbb{R})$ we have that

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k \text{ is absolutely convergent.}$$

How to compute e^A ??? (for the moment, with MAPLE).

Theorem. For $X' = AX$, the principal matrix solution is $E(t) = e^{tA}$, $t \in \mathbb{R}$.

Thus, for any $\eta \in \mathbb{R}^2$ the sol. of the IVP $X' = AX, X(0) = \eta$ is $\varphi(t, \eta) = e^{tA}\eta$.

Exercise. For $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$, compute e^{tD} using the definition of the exponential matrix.

$$e^{tD} = \sum_{k=0}^{\infty} \frac{1}{k!} (tD)^k = \sum_{k=0}^{\infty} \frac{t^k}{k!} D^k$$

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \Rightarrow D^2 = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \begin{pmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{pmatrix}$$

induction: $D^k = \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix}, \quad \forall k \in \mathbb{N}$

$$\Rightarrow e^{tD} = \sum_{k=0}^{\infty} \begin{pmatrix} \frac{t^k \lambda_1^k}{k!} & 0 \\ 0 & \frac{t^k \lambda_2^k}{k!} \end{pmatrix} = \begin{pmatrix} \sum_{k=0}^{\infty} \frac{1}{k!} (\lambda_1 t)^k & 0 \\ 0 & \sum_{k=0}^{\infty} \frac{1}{k!} (\lambda_2 t)^k \end{pmatrix} =$$

$$= \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}.$$

lemma. $\lambda_1, \lambda_2 \in \mathbb{R}.$ $e^{t \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}} = \begin{pmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_2 t} \end{pmatrix}, \quad t \in \mathbb{R}.$

lemma. Let $A, B, P \in \mathcal{M}_2(\mathbb{R})$. Assume that P is invertible and

$$A = P B P^{-1}.$$

Then $e^{tA} = P e^{tB} P^{-1}.$